Machine Learning Project: To bee or not to bee

Team: Lianghong LI, Zekai YAN, Muzi LI

Contributions

- Lianghong LI: project lead; implemented feature extraction, model training, validation, final prediction pipeline, and report integration.
- Zekai YAN: support contribution for data checking, visualization review, and report proofreading.
- Muzi LI: support contribution for reproducibility checking, documentation review, and result verification.

Dataset and Data Issue

Training labels are normalized to ID, bug type, and species. The supervised target is bug type; species is used only for visualization. ID 154 was excluded because train/masks/binary_154.tif is unavailable. No artificial mask was created, estimated, manually drawn, or imputed. The effective training set is 249 samples.

Feature Extraction

Features include color symmetry, shape symmetry, bug pixel ratio, RGB statistics inside the mask, morphology, Hu moments, HSV/LAB statistics, grayscale summaries, and edge density.

Supervised Models

Model comparison uses frequent-class stratified cross-validation because singleton classes make full-label stratified CV invalid. The selected model is refit on all 249 usable samples.

model	accuracy	macro_f1	weighted_f1	status
mlp	0.871	0.755	0.868	ok
svm_rbf	0.863	0.754	0.859	ok
logistic_regression	0.851	0.698	0.856	ok
extra_trees	0.847	0.618	0.827	ok
gradient_boosting	0.798	0.597	0.790	ok
knn	0.786	0.583	0.766	ok
random_forest	0.819	0.564	0.796	ok
dummy_most_frequent	0.464	0.127	0.294	ok

Selected model: mlp. Validation macro-F1: 0.7553521749357212.

Clustering

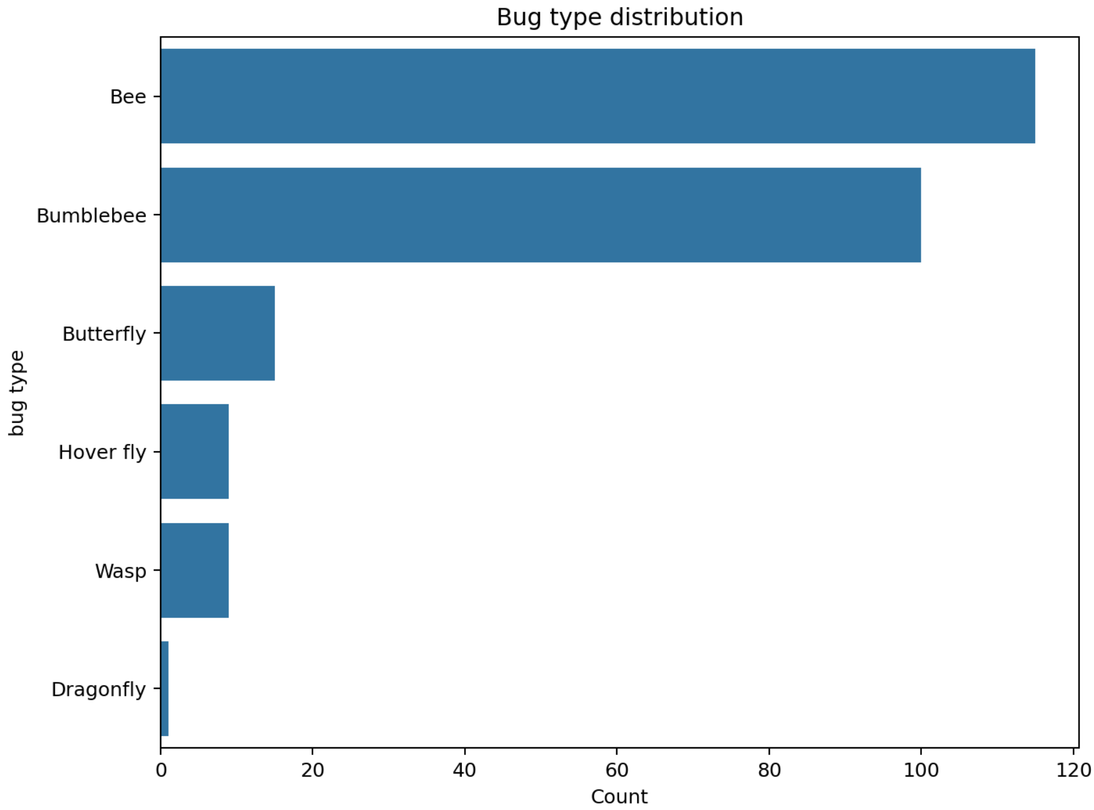
method	cluster_count	silhouette	adjusted_rand_index	normalized_mutual_information
kmeans	6.000	0.113	0.149	0.235
agglomerative	6.000	0.077	0.076	0.181
dbscan	0.000		0.000	0.000

Final Prediction Pipeline

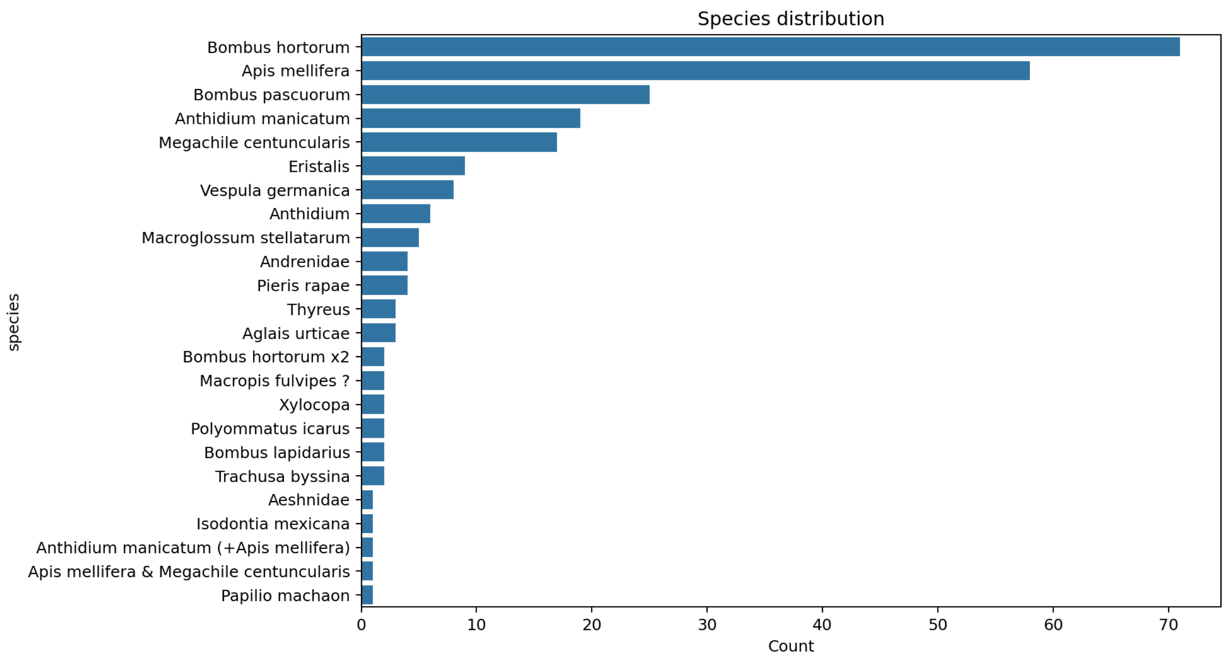
The final submission generation pipeline has been implemented, but the official CSV for images 251–347 will be generated after the test images and masks are released.

Generated Figures

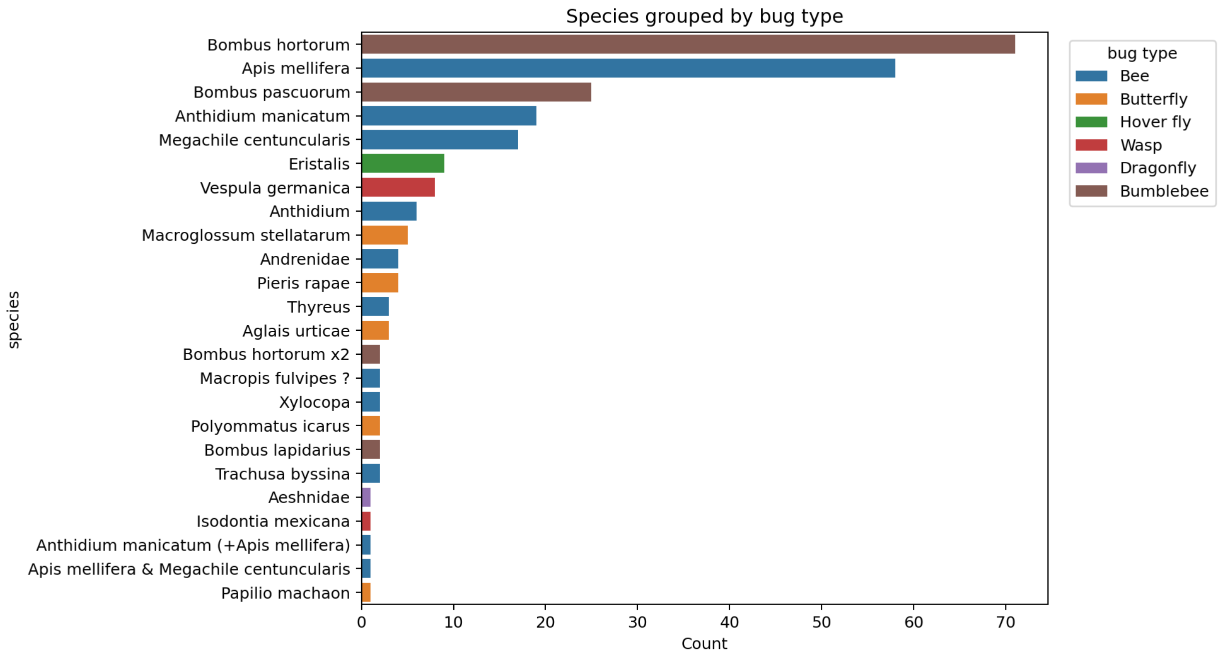
Bug type distribution



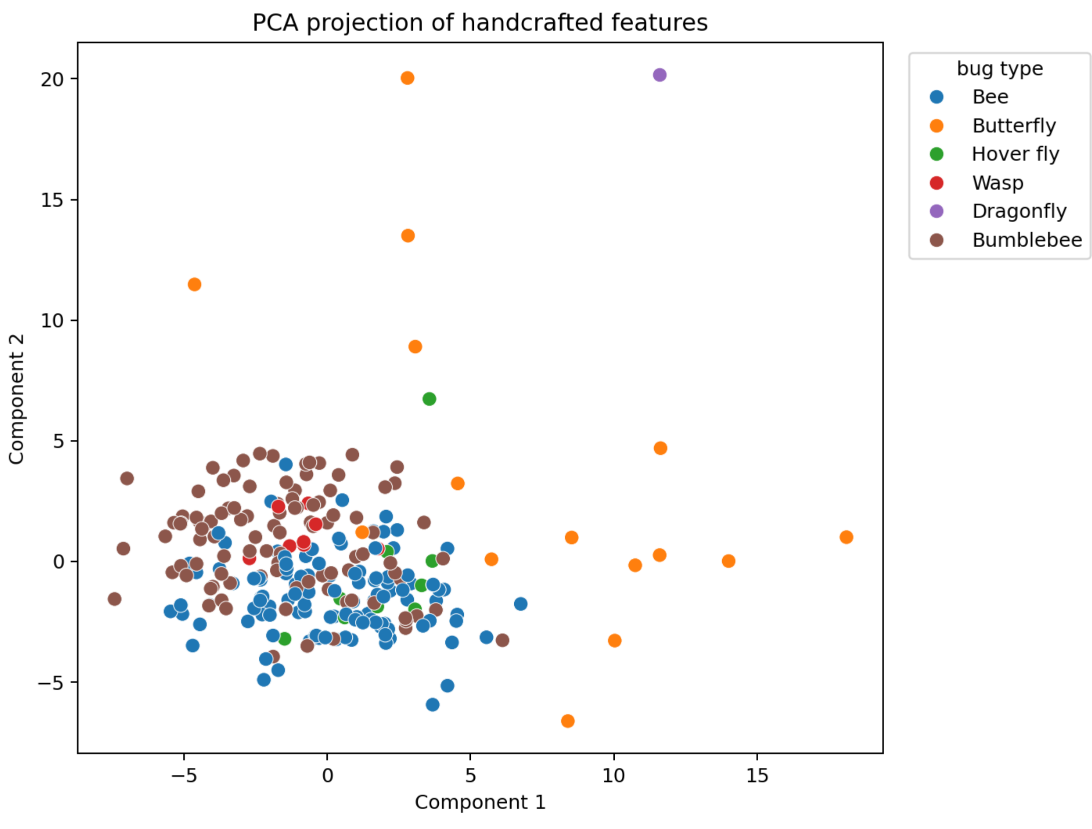
Species distribution



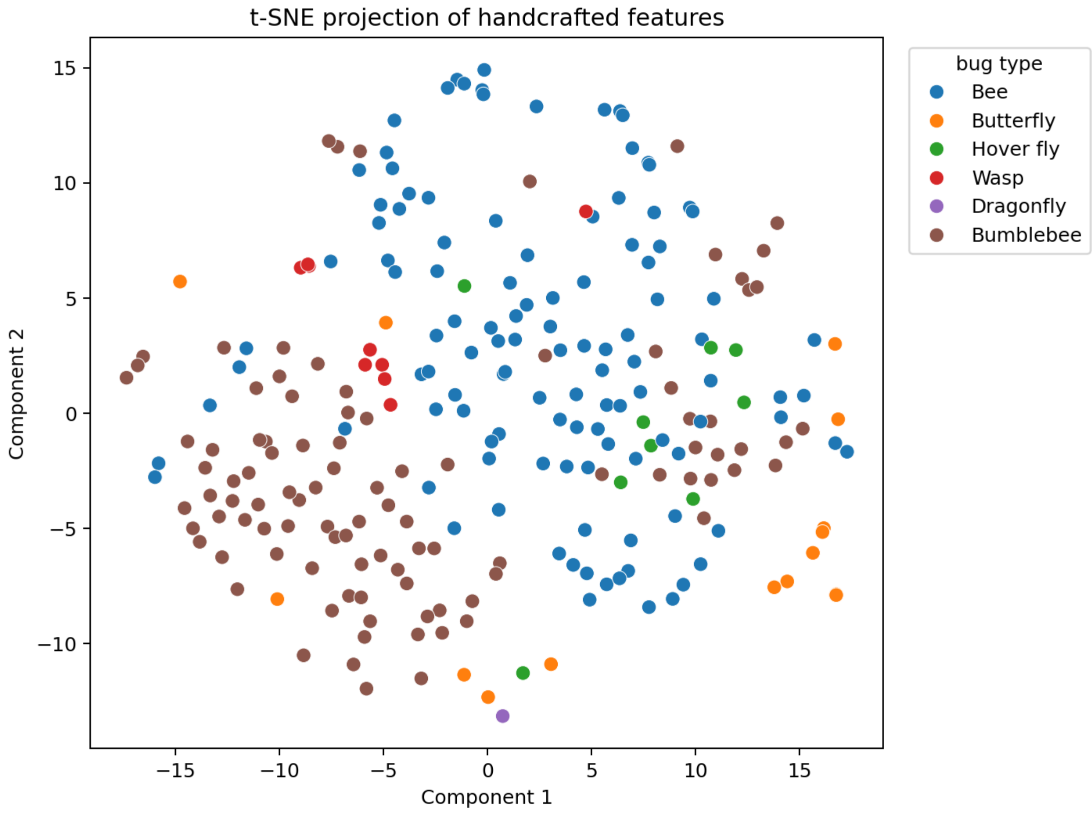
Species grouped by bug type



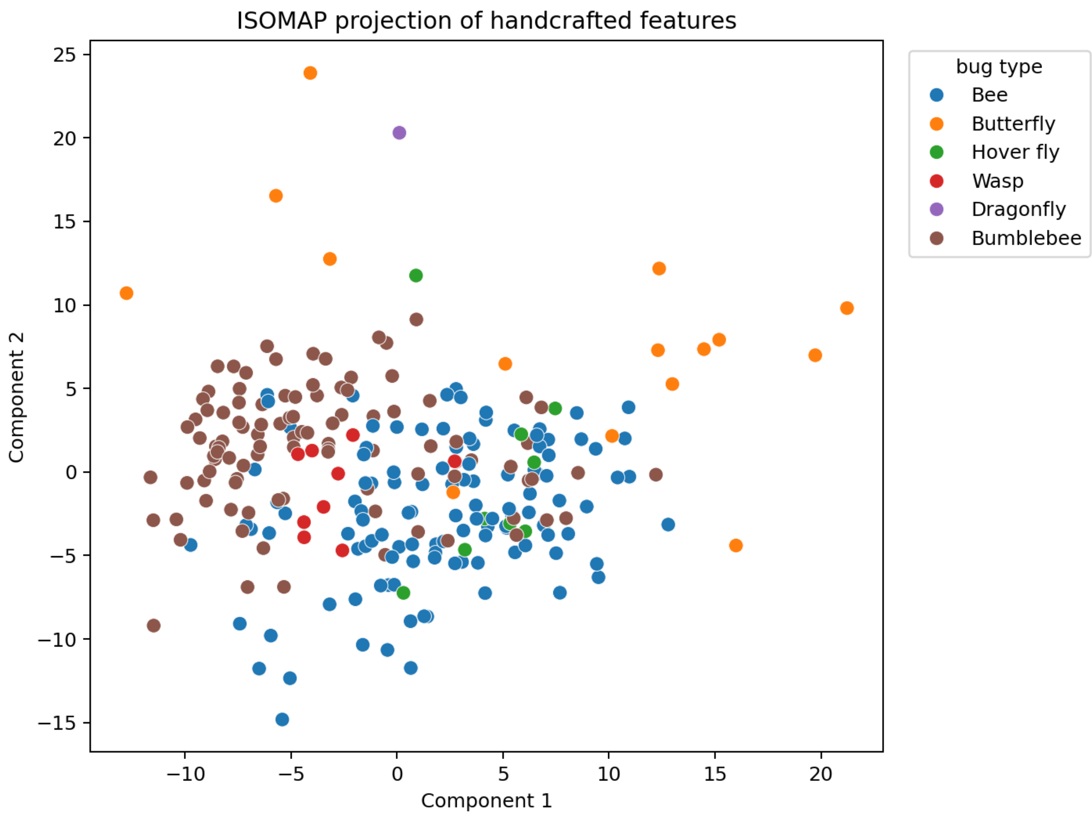
PCA projection



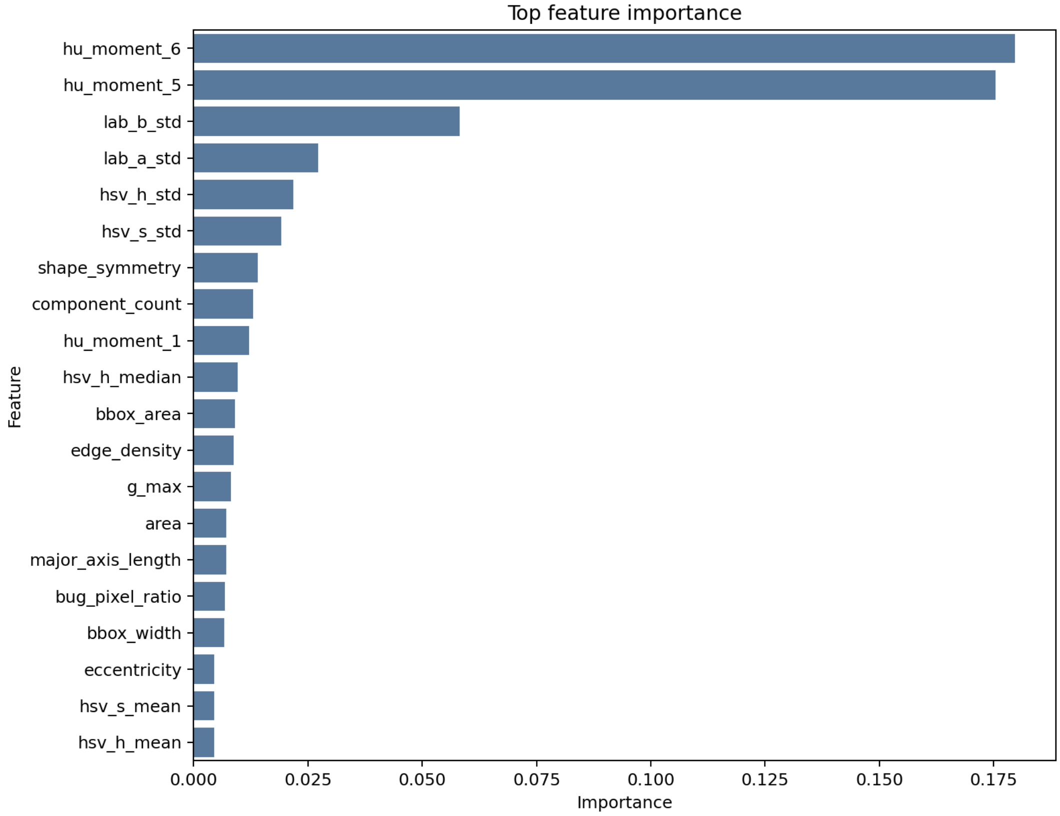
t-SNE projection



Nonlinear projection



Feature importance



Reproducibility

Run compileall, pytest, train validation, feature extraction, model training, clustering, visualization, and report generation from the project root. Do not run final prediction until official test images and masks for IDs 251-347 are available.